Fluid Dynamics of the Atmosphere and Ocean

Problem Set 5

Unassessed.

1. The linearized Boussinesq momentum equations are

$$\frac{\partial u'}{\partial t} = -\frac{\partial \phi'}{\partial x}, \qquad \frac{\partial w'}{\partial t} = -\frac{\partial \phi'}{\partial z} + b',$$

and the linearized mass continuity and thermodynamic equations are

$$\frac{\partial u'}{\partial x} + \frac{\partial w'}{\partial z} = 0, \qquad \frac{\partial b'}{\partial t} + w'N^2 = 0,$$

where N^2 is a constant. There are no variations in the y-direction.

(a) Show that these equations can be reduced to a single equation for w, namely

$$\left[\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2}\right)\frac{\partial^2}{\partial t^2} + N^2\frac{\partial^2}{\partial x^2}\right] w' = 0.$$

[Hint: One possible way to begin is by differentiating the mass continuity equation with respect to time.]

(b) By assuming solutions of the form $w' = \mathrm{Re}\, W \exp[\mathrm{i}(kx + mz - \omega t)]$ obtain the dispersion relation:

$$\omega^2 = \frac{k^2 N^2}{k^2 + m^2}.$$

What kind of waves are these?

(c) Show that the vertical components of the group velocity and the phase speed have opposite signs.

(d) If there is a vertically localized source of these waves, do the wave-crests move toward or away from the wave source?

2. Consider a layer of fluid of constant density at in the upper ocean that satisfies the *Ekman-layer* equations:

$$-fv = -\frac{\partial \phi}{\partial x} + \frac{\partial \tau_x}{\partial z}, \qquad fu = -\frac{\partial \phi}{\partial y} + \frac{\partial \tau_y}{\partial z}.$$

where τ_x, τ_y are components of the stress, $\boldsymbol{\tau}$, in the x- and y-directions and $f = f_0 + \beta y$. Assume that the pressure, ϕ, is not a function of z, that the Ekman layer has some finite depth, H_E, below which the stress is zero, and that the vertical velocity is zero at the top of the ocean, $z = 0$, and at the bottom, $z = -H_o$.

(a) Show that the total ageostrophic transport induced by the stress is at right angles to the direction of the surface stress.

(b) By integrating the mass continuity equation over the entire depth of the Ekman layer show that the vertically integrated geostrophic velocity is given by

$$\int_{-H_o}^{0} \beta v_g \,\mathrm{d}z = f\left[\frac{\partial}{\partial x}\left(\frac{\tau_{y0}}{f}\right) - \frac{\partial}{\partial y}\left(\frac{\tau_{x0}}{f}\right)\right]. \tag{A}$$

(c) By cross-differentiating equations (E) and vertically integrating over the total depth of the ocean, or otherwise, derive the Sverdrup relation,

$$\int_{-H_o}^{0} \beta v \,\mathrm{d}z = \frac{\partial \tau_{y0}}{\partial x} - \frac{\partial \tau_{x0}}{\partial y} \tag{B}$$

where v is the meridional component of the total velocity.

(d) If f is not constant eq. (A) is different from eq. (B). Explain how and why the two equations are nevertheless consistent.